

Optimizing the parity filter by a null polynomial

Research note, 30 September 2026

Theorem 1. *If rational numbers $x_1, \dots, x_K$ satisfy*

$$\frac{1}{K} \sum_{i=1}^K x_i^j = \frac{1}{j+1} \quad (0 \leq j \leq r),$$

then

$$K \geq 2^{2\lfloor r/2 \rfloor}.$$

The nodes may repeat and need not belong to $[0, 1]$.

Write $v = v_2$, and recall from `../rational_designs/two_adic_lower_bound.tex` that

$$G(z) = \frac{z}{\log(1+z)} = \sum_{j \geq 0} g_j z^j, \quad v(g_j) = -j.$$

For completeness, $G(-2t)^{-1} = \sum_{j \geq 0} 2^j t^j / (j+1)$ has integral coefficients and is $1+t$ modulo 2. Thus $G(-2t) \equiv (1+t)^{-1} \pmod{2}$, proving the valuation assertion. Also $g_j = \int_0^1 \binom{x}{j} dx$.

Lemma 2. *Let $N = 2^a$ with $a \geq 1$, and put*

$$I_j(N) = \int_0^1 \binom{Nx}{j} dx.$$

Then

$$v(I_j(N)) \geq -\lfloor j/N \rfloor, \quad v(I_{N\ell}(N)) = -\ell \quad (\ell \geq 0).$$

Proof. Coefficientwise integration gives

$$\sum_{j \geq 0} I_j(N) z^j = G(W(z)), \quad W(z) = (1+z)^N - 1.$$

Every coefficient of W below degree N is even, whereas its leading coefficient is one. Consider a contribution from $g_k W^k$ to degree j , and let b be the number of its factors having degree less than N . It has valuation at least $b-k$. If $L = \lfloor j/N \rfloor$, at most L factors can have degree N , hence $k-b \leq L$. This proves the bound.

For $j = N\ell$, no term with $k < \ell$ occurs. The term with $k = \ell$ is uniquely $g_\ell z^{N\ell}$ and has valuation $-\ell$. If $k > \ell$, having ℓ factors of degree N already leaves no room for the other positive-degree factors. Thus $k-b \leq \ell-1$, and all these contributions have valuation at least $1-\ell$. They cannot cancel the unique term of valuation $-\ell$. $\square$

The classical parity filter is

$$H_s(y) = \sum_{j=1}^s (-2)^{j-1} \binom{y}{j}.$$

For $y \in \mathbb{Z}_2$, the binomial expansion of $(-1)^y$, first for nonnegative integers and then by continuity, gives

$$H_s(y) \equiv \mathbf{1}_{y \notin 2\mathbb{Z}_2} \pmod{2^s}.$$

This filter is attributed to Hua in Mit'kin, Lemma 7 and the following remark; see the historical discussion and primary source in the note cited above. The modification below is the additional step in this derivation; its novelty in the literature has not been audited.

Lemma 3. *For every even $s \geq 2$ and every $N = 2^a$, $a \geq 1$, there is a rational polynomial P of degree at most s satisfying*

$$P(y) \equiv \mathbf{1}_{y \notin 2\mathbb{Z}_2} \pmod{2^s} \quad (y \in \mathbb{Z}_2), \quad \int_0^1 P(Nx) dx = 0.$$

Proof. Let $J = \int_0^1 H_s(Nx) dx$ and $L = \lfloor s/N \rfloor$. By Lemma 2, the coefficients $A_j = (-2)^j I_j(N)$ obey

$$v(A_j) \geq j - \lfloor j/N \rfloor.$$

The series converges at $z = -2$. Moreover, $W(-2) = (-1)^N - 1 = 0$, so

$$\sum_{j \geq 0} A_j = G(W(-2)) = 1.$$

This evaluation is legitimate directly from the coefficient bound: the contributions to degree j in the formal composition have the same lower bound, which tends to infinity. Hence regrouping is valid. Consequently

$$J = -\frac{1}{2} \sum_{j=1}^s A_j = \frac{1}{2} \sum_{j>s} A_j, \quad v(J) \geq s - \lfloor (s+1)/N \rfloor = s - L.$$

The last equality uses that s and N are even, so N cannot divide $s+1$. Put $j_0 = NL \leq s$; this includes $j_0 = 0$ if $N > s$. Lemma 2 gives $v(I_{j_0}(N)) = -L$, so the rational number

$$t = -\frac{J}{2^s I_{j_0}(N)}$$

belongs to $\mathbb{Z}_2$. Now set

$$P(y) = H_s(y) + 2^s t \binom{y}{j_0}.$$

The added polynomial takes values in $2^s \mathbb{Z}_2$ on $\mathbb{Z}_2$, and by construction its integral cancels J exactly. $\square$

Proof of Theorem 1. Take $s = 2\lfloor r/2 \rfloor$; the case $s = 0$ is immediate. If every x_i is 2-adically integral, exactness for $\binom{x}{s}$ and $v(g_s) = -s$ imply $2^s \mid K$. Otherwise choose the least $a \geq 1$ for which every $y_i = 2^a x_i$ lies in $\mathbb{Z}_2$. At least one y_i is odd. For the polynomial in Lemma 3, exactness gives

$$\sum_{i=1}^K P(y_i) = K \int_0^1 P(2^a x) dx = 0.$$

The number of odd y_i is therefore a positive multiple of 2^s . It is at most K , proving the theorem. $\square$

Corollary 4 (Equality forces odd denominators). *If $r \geq 2$ is even and $K = 2^r$, then every node belongs to $\mathbb{Z}_2$. Equivalently, each node has odd denominator in lowest terms.*

Proof. Otherwise normalize by $N = 2^a$ as before. The positive unit count is a multiple of $2^r = K$, so all $y_i = Nx_i$ are odd. Consequently $z_i = (y_i - 1)/2 \in \mathbb{Z}_2$. Put $M = 2^{a-1}$. We claim

$$v\left(\int_0^1 \binom{Mx - 1/2}{r} dx\right) = -r - v(r!).$$

For $a \geq 2$, expansion around $-1/2$ gives, for every $j \geq 0$,

$$\int_0^1 (Mx - 1/2)^j dx = (-1/2)^j \sum_{\ell=0}^j \binom{j}{\ell} \frac{(-2M)^\ell}{\ell + 1}.$$

Every summand with $\ell > 0$ inside the sum is even, since $a\ell - v(\ell + 1) \geq 1$. Thus this integral has valuation $-j$. For $a = 1$, the odd moments vanish and the even moments have valuation $-j$, by symmetry about $1/2$. As r is even, the monic degree- r numerator of $\binom{u}{r}$ therefore has a unique lowest-valuation integrated term, namely u^r . This proves the claim. Exactness and the integrality of every $\binom{z_i}{r}$ now force $v(K) \geq r + v(r!) > r$, contradicting $K = 2^r$. $\square$

Corollary 5 (An odd-degree refinement). *For odd $r \geq 3$, let o be the odd part of $r + 1$. Then*

$$K \geq 2^{r-1} + 2^{o-1}.$$

In particular, degree 5 requires $K \geq 20$, and degree 9 requires $K \geq 272$.

Proof. If all nodes lie in $\mathbb{Z}_2$, then $2^r \mid K$, which is stronger. Otherwise take the same minimal normalization $N = 2^a$ and let U be the positive number of odd normalized nodes. Assume $K < 2^r$. The degree- $(r - 1)$ even filter shows that $U = 2^{r-1}$.

Write $J = \int_0^1 H_r(Nx) dx$. If $N \nmid r + 1$, then $\lfloor (r + 1)/N \rfloor = \lfloor r/N \rfloor$; the proof of Lemma 3 works unchanged in degree r and gives $2^r \mid U$, a contradiction. Thus $N \mid r + 1$; put $q = (r + 1)/N$. The first term in the tail representation of J is $A_{r+1}/2$, with valuation exactly $r - q$ by Lemma 2. Every later term has strictly greater valuation, since $j - \lfloor j/N \rfloor > (r + 1) - q$ for $j > r + 1$. Hence $v(J) = r - q$. Exactness and the parity congruence give

$$U \equiv KJ \pmod{2^r}.$$

As $v(U) = r - 1$, necessarily $v(K) = q - 1$. We have $q \geq o$ and $q - 1 \leq (r - 1)/2 < r - 1$. Thus K is strictly larger than 2^{r-1} , and the next integer above that value with valuation $q - 1$ is $2^{r-1} + 2^{q-1}$. This proves the stated bound. $\square$

Corollary 6 (The exact minimum in degree five). *The minimum number of nodes in a rational equal-weight uniform-interval quadrature of degree five is exactly 20.*

Proof. Corollary 5 gives the lower bound 20. For the upper bound, take the following multiset of ten magnitudes:

$$(s_1, \dots, s_{10}) = (1, 11, 2, 2, 6, 6, 8, 12, 12, 14),$$

and the twenty rational nodes

$$x_{i,\pm} = \frac{15 \pm s_i}{30} \quad (1 \leq i \leq 10).$$

They all lie strictly between zero and one. On the centered interval $[-1, 1]$, these nodes are $\pm s_i/15$. Their odd moments vanish by symmetry, while

$$\sum_i s_i^2 = 750, \quad \sum_i s_i^4 = 101250$$

give the second and fourth moments $1/3$ and $1/5$ exactly. The affine change of variables proves exactness through degree five. $\square$

Example and scope. For $s = 2, N = 2$, one may take $P(y) = 4y - 3y^2$. Its values are the parity indicator modulo 4, and $\int_0^1 P(2x) dx = 0$. The theorem concerns rational scalar quadrature; it does not supply an exponential lower bound for arbitrary permutation-fair dice.